SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence   COURSE CODE: CSA1703

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

Time Duration: 1 Hour

QUESTIONS: 5   Total Marks:30

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I E.Ragul / 192571032 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

E.Ragul

Annexure A

Resolution Algorithm Implementation

Question 1 – Rain and Wet Ground
Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that the ground is wet **using the Resolution Algorithm.**

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF.

Negate the goal.

1. Apply the Resolution Algorithm step by step.
2. Show the Empty Clause (□) if the goal is proved.

ANSWER:

Resolution Algorithm – Rain and Wet Ground
1. Represent the statements in Propositional Logic

Let:

- R = It is raining.
- W = The ground is wet.

Knowledge Base:

1. If it rains, then the ground becomes wet.
   $R \rightarrow W$
2. It is raining.
   R

Goal:

- W = The ground is wet.

2. Convert the statements into CNF

The implication:

$R \rightarrow W$

is equivalent to:

$\neg R \lor W$

So the CNF clauses are:

- Clause 1: $\neg R \lor W$
- Clause 2: R

3. Negate the Goal

Goal:

W

Negated goal:

$\neg W$

Now the clauses are:

1. $\neg R \lor W$
2. R
3. $\neg W$

4. Apply Resolution Algorithm
Step 1: Resolve Clause 1 and Clause 2

Clauses:

- $\neg R \lor W$
- R

The complementary literals are $\neg R$ and R.

After resolution:

W

So we obtain:

Clause 4: W

Step 2: Resolve Clause 4 and Clause 3

Clauses:

- W
- ¬W

The complementary literals are W and ¬W.

After resolution:

□

The symbol □ is the Empty Clause.

5. Resolution Proof
1. ¬R ∨ W (Rain implies wet ground)
2. R (It is raining)
3. ¬W (Negation of goal)

Resolve 1 and 2:
4. W

Resolve 4 and 3:
5. □

Since the Empty Clause (□) is obtained, the negation of the goal leads to a contradiction.

Therefore:

∴ W is proved
∴ The ground is wet.

6. Final Conclusion

The Resolution Algorithm proves that the ground is wet from the knowledge base. This demonstrates logical reasoning using propositional logic, CNF conversion, negation of the goal, and resolution. The empty clause confirms that the goal logically follows from the given facts.

Question 2 – Student Assignment Submission
Knowledge Base

1. If a student submits the assignment, then the student receives internal marks
2. Rahul submitted the assignment.

Goal

Prove that Ragul receives internal marks **using the Resolution Algorithm.**

Tasks
1. Write the propositional logic expressions.
2. Convert the premises into CNF.

Negate the goal.

1. Apply the Resolution Algorithm.
2. Conclude whether the goal is proved.

ANSWER:

1. Propositional Logic

Let

- S = Ragul submitted the assignment.
- M = Ragul receives internal marks.

Knowledge Base:

1. If Ragul submits the assignment, then Rahul receives internal marks.
   $S \rightarrow M$
2. Ragul submitted the assignment.
   S

Goal:

M = Rahul receives internal marks.

2. CNF Clauses

Convert:

$S \rightarrow M$

to:

$\neg S \lor M$

Therefore:

- Clause 1: ¬S ∨ M
- Clause 2: S

3. Negate the Goal

Goal:

M

Negated goal:

¬M

So the clauses are:

1. ¬S ∨ M
2. S
3. ¬M

4. Resolution Steps

Step 1: Resolve Clause 1 and Clause 2.

¬S ∨ M
S
M

New clause:

Clause 4: M

Step 2: Resolve Clause 4 and Clause 3.

M
¬M

New clause:

Clause 5: □

5. Resolution Tree

```
   ¬S ∨ M     S
      \      /
       \    /
         M
         |
        ¬M
```

Question 3 – Library Membership
Knowledge Base
1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.
Goal

1. Prove that Priya can borrow books **using the Resolution Algorithm.**

Tasks
1. Represent the knowledge base using propositional logic.
2. Convert each statement into CNF.

Negate the goal.

1. Perform resolution until no new clauses can be generated.
2. State the final conclusion.

ANSWER:

Question 3 – Library Membership
1. Propositional Logic

Let:

- M = Priya is a library member.
- B = Priya can borrow books.

Knowledge Base:

1. If a person is a library member, then the person can borrow books.
   $M \rightarrow B$
2. Priya is a library member.
   M

Goal:

B = Priya can borrow books.

2. Convert Each Statement into CNF

Statement 1:

$M \rightarrow B$

Convert implication:

$\neg M \lor B$

So:

- Clause 1: $\neg M \lor B$

- Clause 2: M

3. Negate the Goal

Goal:

B

Negated goal:

¬B

Now the clauses are:

1. ¬M ∨ B
2. M
3. ¬B

4. Perform Resolution
Step 1: Resolve Clause 1 and Clause 2
¬M ∨ B
M

B

The complementary literals ¬M and M cancel.

Therefore:

Clause 4: B

Step 2: Resolve Clause 4 and Clause 3
B
¬B

□

The complementary literals B and ¬B cancel.

Therefore:

Clause 5: □

The Empty Clause (□) is obtained.

5. Resolution Proof
1. ¬M ∨ B (Library member → can borrow books)
2. M (Priya is a library member)
3. ¬B (Negation of goal)

Resolve 1 and 2:
4. B

Resolve 4 and 3:
5. □
6. Final Conclusion

Since the Empty Clause (□) is obtained, the negation of the goal leads to a contradiction.

Therefore, the goal is proved.

∴ Priya can borrow books.

Question 4 – Placement Eligibility
Knowledge Base
1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.
Goal
1. Prove that Arun is eligible for placement **using the Resolution Algorithm.**
Tasks
1. Convert the statements into propositional logic.
2. Write the CNF clauses.
3. Negate the goal.
4. Apply the Resolution Algorithm step by step.
5. Draw the resolution tree and conclude.

ANSWER:

1. Convert the Statements into Propositional Logic

Let:

- A = Arun cleared the aptitude test.
- E = Arun is eligible for placement.

Knowledge Base:

1. If Arun clears the aptitude test, then Arun is eligible for placement.
   $A \rightarrow E$
2. Arun cleared the aptitude test.
   A

Goal:

E = Arun is eligible for placement.

2. Write the CNF Clauses

Convert the implication:

$A \rightarrow E$

Using:

$P \rightarrow Q = \neg P \lor Q$

Therefore:

$\neg A \lor E$

CNF clauses:

- Clause 1: $\neg A \lor E$
- Clause 2: A

3. Negate the Goal

Goal:

E

Negated goal:

$\neg E$

Now the clauses are:

1. $\neg A \lor E$
2. A
3. $\neg E$

4. Apply Resolution Algorithm Step by Step
Step 1: Resolve Clause 1 and Clause 2
$\neg A \lor E$
A
E

The complementary literals are $\neg A$ and A.

Therefore:

Clause 4: E

Step 2: Resolve Clause 4 and Clause 3
E
$\neg E$

The complementary literals are E and $\neg E$.

Therefore:

Clause 5: □

The Empty Clause (□) is obtained.

5. Resolution Tree

```
           ¬E
            |
            |
        E    ¬E
        |
     ¬A ∨ E
        |
        A
        |
        E
        |
        └──────────── □
```

A clearer resolution tree:

```
            ¬A ∨ E        A
               \         /
                \       /
                   E
                   |
                   |   ¬E
                   \   /
                    \ /
                     □
```

Resolution Steps
1. ¬A ∨ E   Premise
2. A   Premise
3. ¬E   Negated Goal

Resolve (1) and (2):
4. E

Resolve (4) and (3):
5. □

Final Conclusion

Since the Empty Clause (□) is obtained, the negation of the goal produces a contradiction.

Therefore, the goal is proved.

∴ Arun is eligible for placement.

Question 5 – Access Control System
Knowledge Base
1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.
Goal

Prove that the user is granted access **using the Resolution Algorithm.**

Tasks
1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause (□).
5. Explain each resolution step and write the final conclusion.

ANSWER:

1. Propositional Logic

Let:

- P = User entered the correct password.
- A = User is authenticated.
- G = User is granted access.

Knowledge Base:

1. If the user enters the correct password, then the user is authenticated.
   $P \rightarrow A$
2. If the user is authenticated, then the user is granted access.
   $A \rightarrow G$
3. The user entered the correct password.
   P

Goal:

G = User is granted access.

2. Convert All Implications into CNF
Statement 1:

$P \rightarrow A$

Convert implication:

$\neg P \lor A$

Clause 1: $\neg P \lor A$

Statement 2:

$A \rightarrow G$

Convert implication:

$\neg A \vee G$

Clause 2: $\neg A \vee G$

Statement 3:

P

Clause 3: P

3. Negate the Goal

Goal:

G

Negated goal:

$\neg G$

Therefore, the complete set of clauses is:

1. $\neg P \vee A$
2. $\neg A \vee G$
3. P
4. $\neg G$

4. Apply Resolution Algorithm
Step 1: Resolve Clause 1 and Clause 3
$\neg P \vee A$
P
A

The complementary literals are $\neg P$ and P.

Clause 5: A

Step 2: Resolve Clause 2 and Clause 5
$\neg A \vee G$
A
G

The complementary literals are $\neg A$ and A.

Therefore:

Clause 6: G

Step 3: Resolve Clause 6 and Clause 4
G
¬G

The complementary literals are G and ¬G.

Clause 7: □

The Empty Clause (□) is obtained.

5. Resolution Tree

```
¬P ∨ A      P
    \      /
     \    /
       A
       |
       |
¬A ∨ G
    \
     \
      G
      |
     ¬G
      |
      □
```

Resolution Proof
1. ¬P ∨ A Premise
2. ¬A ∨ G Premise
3. P Premise
4. ¬G Negated Goal

Resolve (1) and (3):
5. A

Resolve (2) and (5):
6. G

Resolve (6) and (4):
7. □

Final Conclusion

Since the Empty Clause (□) is obtained, the negation of the goal leads to a contradiction.

Therefore, the goal is proved.

∴ The user is granted access.

RUBRICS FOR CALCULATION

Numerical Problems					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation